

# RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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## Hard examples and a live model

This report adds 24 harder files and a live model. It continues report 6.

### What we did

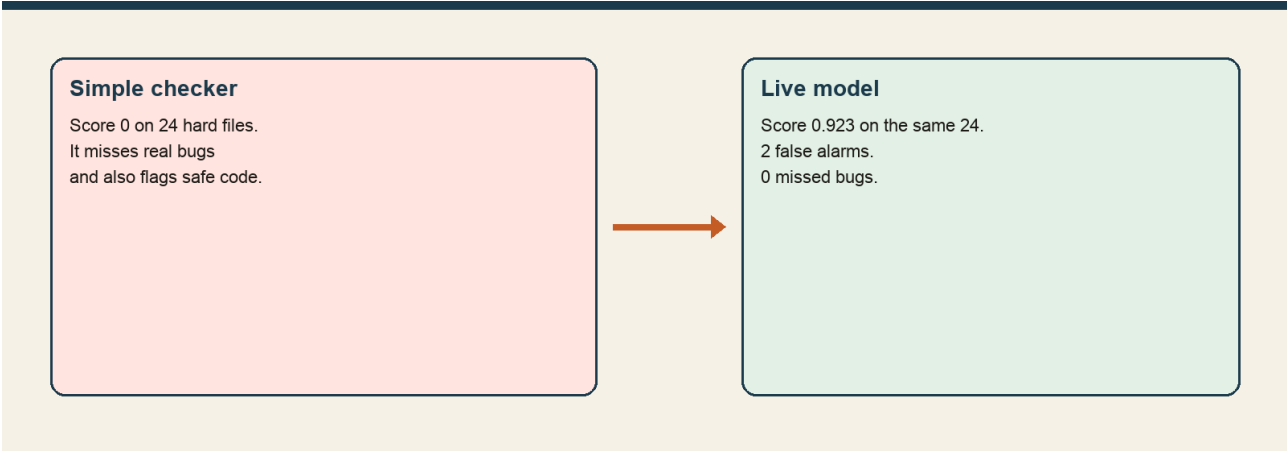
We wrote 24 extra Java files to fool the simple checker. Some safe files look risky. Some unsafe files look quiet.

On these 24, the simple checker scores  $F1 = 0$ . It raises 12 false alarms and misses 12 bugs. The template also scores 0. It sees fewer false alarms only because comments were blanked. It still misses every bug.

The live model is Groq, model openai/gpt-oss-20b. On the same 24 files its F1 is 0.923. It has 2 false alarms and misses none.

We also turned the CWE search off and ran the model again. The score stayed 0.923. So on these 24 files, search did not improve detection.

### Picture



### Numbers for this step only

| System                | F1    | False alarms | Missed bugs | Files |
|-----------------------|-------|--------------|-------------|-------|
| Simple checker        | 0.000 | 12           | 12          | 24    |
| Template              | 0.000 | 6            | 12          | 24    |
| Live model            | 0.923 | 2            | 0           | 24    |
| Live model, no search | 0.923 | 2            | 0           | 24    |

### What is next

Next we will run the simple checker on six public Java test sets.

### **What this report does not claim**

These 24 files were written by us. They are not Juliet. We do not say that CWE search helps detection.